RoboCore ultrasonic sensor: CS100A + 40 kHz transducer pair, HC-SR04 timing. JST SH 4 to RoboCore V2 J403.

Ultrasonic

CS100A single-chip ranger
HC-SR04 TRIG/ECHO, 3.3 V

File: ultrasonic.kicad_sch

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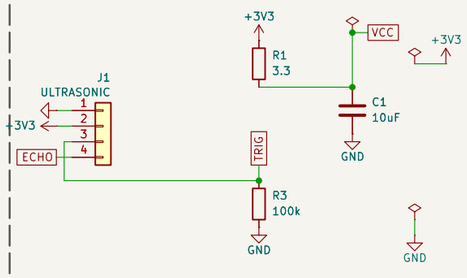
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File: RoboCore_Ultrasonic.kicad_sch

Title: RoboCore Ultrasonic

Size: A4	Date: 2026-10-05	**Rev: 1.0**
KiCad E.D.A. 10.0.0		Id: 1/2

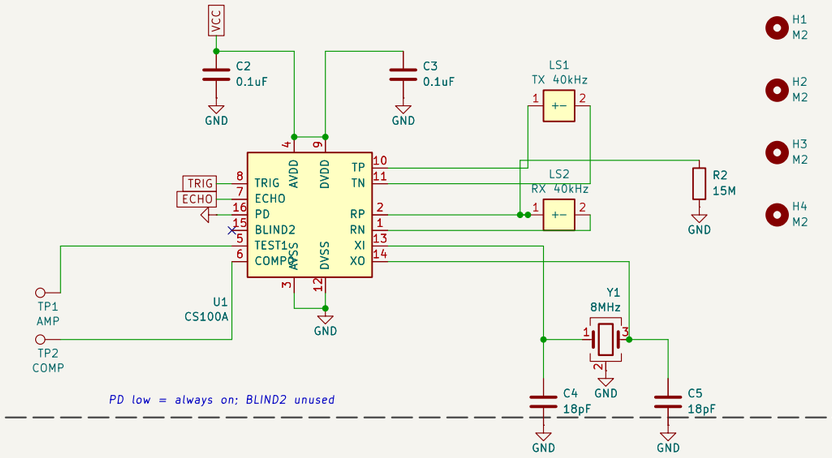
CONNECTOR + SUPPLY FILTER

JST SH 4-pin to the RoboCore V2 (J403): GND, 3V3, TRIG, ECHO (Qwiic power order).
R1/C1 keep the transmitter's current pulses off the robot's 3.3 V rail (CS100A reference circuit).



RANGING (CS100A)

Y1 8 MHz crystal (12 pF load, 18 pF caps). R2 15M sets the receiver sensitivity: lower it (5.1M–22M) if far readings jump around; raise it for more range. TP1/TP2: amplifier and comparator outputs, for tuning.



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Title: RoboCore Ultrasonic – Ultrasonic

Size: A3	Date: 2026-10-05	**Rev: 1.0**
KiCad E.D.A. 10.0.0		Id: 2/2